

09/30/2026: Binomial Coefficients

CSCI 246: Discrete Structures

Textbook reference: Sec 17, Scheinerman

Today's Agenda

- Mini-lecture (30 mins)
- Group exercises (20 mins)

Study Guide For Quiz 5

Group exercises. Know how to do all of the group exercises from the course meetings on Relations (9/23), Equivalence Relations (9/25), and Partitions (9/28).

Readings. Sec. 14 (Relations)

- Understand the definition of relations in Def 14.2.
- Know how to determine whether a relation is reflexive, symmetric, and/or transitive, as in Examples 14.8-14.11. (Skip antisymmetric and irreflexive.)

Sec. 15 (Equivalence Relations)

- Know how to apply the definition of congruence mod n (to determine whether a pair of numbers is congruent).
- Know how to prove that congruence mod n is an equivalence relation.
- Know how to determine equivalence classes as in Examples 15.7 and 15.8.

Sec. 16 (Partitions)

- Understand how partitions and equivalence classes are “two sides of the same coin”.
- Know how to count the number of ways to rearrange the letters in the words WORD, HELLO, and AARDVARK, and justify your answer using equivalence classes.

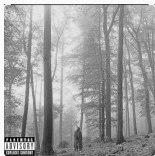
Binomial Coefficients

Five albums

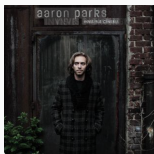
Activity. As we listen to clips from these five albums, think about how you would sort them in terms of how much you like them.



**Kendrick
Lamar**
good kid,
m.A.A.d city



Taylor Swift
folklore



Aaron Parks
Invisible
Cinema



**Chappell
Roan**
The Rise and
Fall of a
Midwest
Princess



Zach Bryan
Zach Bryan

Five albums



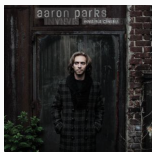
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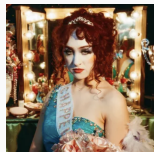
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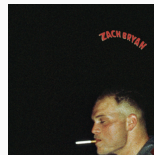
Aaron Parks

Invisible Cinema



Chappell Roan

The Rise and Fall of a
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Zach Bryan

Zach Bryan

Activity. Find a partner and answer these questions:

1. How many ways are there to rank (sort) these five albums?
2. How many ways are there to pick a favorite and second favorite album?
3. How many ways are there to pick 2 albums for a road trip to Helena?

Number of ways to rank the five albums

The number of ways to sort (rank) all five albums is

$$5 \cdot 4 \cdot 3 \cdot 2 \cdot 1.$$

To see this, let's use the shorthand: **Kendrick** **Taylor** **Aaron** **Chappell** **Zach**

K ...	T ...	A ...	C ...	Z ...
KTACZ	TKACZ	AKTCZ	CKTAZ	ZKTAC
KTAZC	TKAZC	AKTZC	CKTZA	ZKTCA
KTCAZ	TKCAZ	AKCTZ	CKATZ	ZKATC
KTCZA	TKCZA	AKCZT	CKAZT	ZKACT
KTZAC	TKZAC	AKZTC	CKZTA	ZKCTA
KTZCA	TKZCA	AKZCT	CKZAT	ZKCAT
KATCZ	TAKCZ	ATKCZ	CTKAZ	ZTKAC
KATZC	TAKZC	ATKZC	CTKZA	ZTKCA
KACTZ	TACKZ	ATCKZ	CTAKZ	ZTAKC
KACZT	TACZK	ATCZK	CTAZK	ZTACK
KAZTC	TAZKC	ATZKC	CTZKA	ZTCKA
KAZCT	TAZCK	ATZCK	CTZAK	ZTCAK
KCTAZ	TCKAZ	ACKTZ	CAKTZ	ZAKTC
KCTZA	TCKZA	ACKZT	CAKZT	ZAKCT
KCATZ	TCAKZ	ACTKZ	CATKZ	ZATKC
KCAZT	TCAZK	ACTZK	CATZK	ZATCK
KCZTA	TCZKA	ACZKT	CAZKT	ZACKT
KCZAT	TCZAK	ACZTK	CAZTK	ZACTK
KZTAC	TZKAC	AZKTC	CZKTA	ZCKTA
KZTCA	TZKCA	AZKCT	CZKAT	ZCKAT
KZATC	TZAKC	AZTKC	CZTKA	ZCTKA
KZACT	TZACK	AZTCK	CZTAK	ZCTAK
KZCTA	TZCKA	AZCKT	CZAKT	ZCAKT
KZCAT	TZCAK	AZCTK	CZATK	ZCATK

Rationale: 5 choices for the first album $\times$ 4 choices for the second album $\times \dots$

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KACZT	TACZK	ATCZK	CTAZK	ZTACK
KAZTC	TAZKC	ATZKC	CTZKA	ZTCKA
KAZCT	TAZCK	ATZCK	CTZAK	ZTCAK
KCTAZ	TCKAZ	ACKTZ	CAKTZ	ZAKTC
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Notation: We write this as $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$.

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Notation: Alternatively, we can write this as $(5)_5 = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$.

Now just pick a #1 and a #2

The number of ways to choose a 1st and 2nd favorite is

$$5 \cdot 4.$$

To see this, let's use the shorthand: **K**endrick **T**aylor **A**aron **C**happell **Z**ach

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Rationale: 5 choices for the favorite album $\times$ 4 choices left for the 2nd favorite album $\times \dots$

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The number of ways to choose two albums is

$$\frac{5 \cdot 4}{2} = 10$$

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Rationale: Matching colors are the **same choice of two albums**:
• {K,Z} • {T,A} • {T,C} • {T,Z} • {A,C} • {A,Z} • {C,Z}. Every class has exactly 2 members (e.g., KT and TK), so $20 \div 2 = 10$ ways to pick two albums.

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Filling k slots from a pool of n items – no item is reused, and **order matters**

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We write this as $\binom{n}{k}$.

Definition of Binomial Coefficient

Let $n, k \in \mathbb{N}$. The symbol $\binom{n}{k}$ denotes the number of k -element subsets of an n -element set.

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Poll - What is the difference between combinations and permutations?

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Poll - What is the difference between combinations and permutations?

- The number of **combinations**, $\binom{n}{k}$, is the number of outcomes when order doesn't matter. We are thinking here in terms of sets.
- The number of **permutations** $(n)_k = n \cdot (n-1) \cdot \dots \cdot (n-k+1)$, is the number of outcomes where order matters. We are thinking here in terms of lists.

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Example

Suppose we're making an ice cream sundae.

- Combinations: The selections $(\text{M\&M's}, \text{fudge}) = (\text{fudge}, \text{M\&M's})$.
- Permutations: The selections $(\text{M\&M's}, \text{fudge}) \neq (\text{fudge}, \text{M\&M's})$.

Scheinerman Prop 17.7

Let $n, k \in \mathbb{N}$ with $0 \leq k \leq n$. Then

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Example. There are as many ways to choose 4 things from 100 as to choose 96 things from 100.

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Symmetry

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Poll: What is the intuition?

One answer. Imagine a class with n children. The teacher has k identical candy bars to give to exactly k children. In how many ways can the k candy bars be distributed?

- The optimist: $\binom{n}{k}$, because we are selecting k lucky children to get the candy.
- The pessimist: $\binom{n}{n-k}$, because we are selecting $n - k$ unlucky children to not get the candy.



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Take home. “Choosing k is the same as leaving out $n - k$.”

Symmetry: Detailed Example

Example

Suppose the class has $n = 5$ students and we select $k = 2$ students to get candy. Let A be a selection of students who get candy, and $\bar{A}$ be the complementary set (i.e. selection of students who get no candy). Now note that

- $\binom{n}{k} = \binom{5}{2}$ is the number of different choices for A .
- $\binom{n}{n-k} = \binom{5}{3}$ is the number of different choices for $\bar{A}$.

But each possible A is paired up with some $\bar{A}$. So the number of choices for each must be equal.

A	$\bar{A}$	A	$\bar{A}$
$\{1, 2\}$	$\{3, 4, 5\}$	$\{2, 4\}$	$\{1, 3, 5\}$
$\{1, 3\}$	$\{2, 4, 5\}$	$\{2, 5\}$	$\{1, 3, 4\}$
$\{1, 4\}$	$\{2, 3, 5\}$	$\{3, 4\}$	$\{1, 2, 5\}$
$\{1, 5\}$	$\{2, 3, 4\}$	$\{3, 5\}$	$\{1, 2, 4\}$
$\{2, 3\}$	$\{1, 4, 5\}$	$\{4, 5\}$	$\{1, 2, 3\}$

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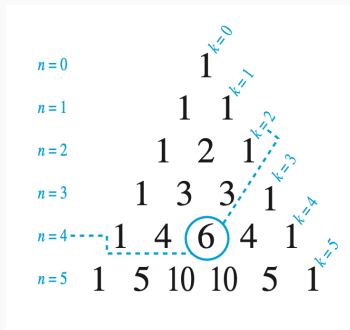
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Pascal's Triangle

$$\begin{array}{ccccccc} & & \binom{0}{0} & & & & \\ & \binom{1}{0} & & \binom{1}{1} & & & \\ & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & \\ & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} \\ & \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4} \\ & \binom{5}{0} & & \binom{5}{1} & & \binom{5}{2} & & \binom{5}{3} & & \binom{5}{4} & & \binom{5}{5} \end{array}$$

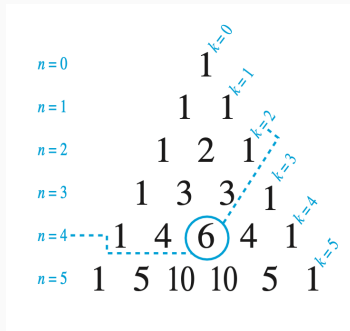
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Pascal's Triangle

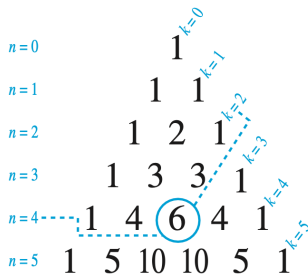
$$\begin{array}{ccccccc} & & \binom{0}{0} & & & & \\ & \binom{1}{0} & & \binom{1}{1} & & & \\ & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & \\ & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} \\ & \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4} \\ \binom{5}{0} & & \binom{5}{1} & & \binom{5}{2} & & \binom{5}{3} & & \binom{5}{4} & & \binom{5}{5} \end{array}$$



Do you notice any pattern?

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$$\begin{array}{ccccccc} & & \binom{0}{0} & & & & \\ & \binom{1}{0} & & \binom{1}{1} & & & \\ & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & \\ & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} \\ & \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4} \\ & \binom{5}{0} & & \binom{5}{1} & & \binom{5}{2} & & \binom{5}{3} & & \binom{5}{4} & & \binom{5}{5} \end{array}$$



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Algorithm for forming Pascal's Triangle

- Form an upside-down V shape of 1's. (Why?)
- The intermediate number in any row is formed by adding the two numbers just to its left and just to its right in the previous row.

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

for integers k and n where $0 < k < n$.

Pascal's Triangle

Pascal's Identity

Let k and n be integers with $0 < k < n$. Then

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Poll: What is the intuition?

Pascal's Triangle

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Poll: What is the intuition?

Pick one of the elements to be “fire”.

- If we include the fire element in our selected subset, we have $\binom{n-1}{k-1}$ ways to complete the subset.
- If we don't include the fire element in our selected subset, then we have $\binom{n-1}{k}$ ways to form the subset.

Binomial Coefficients: Computational Formula

Scheinerman Thm 17.12: Formula for $\binom{n}{k}$

Let k and n be integers with $0 \leq k \leq n$. Then

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Poll: How to understand this formula?

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Poll: How to understand this formula?

1. Suppose we want to select k items from a set of n items, and order matters. Then the number of possible selections is

$$(n)_k = n \cdot (n-1) \cdot \dots \cdot (n-k+1) \quad (\text{From Sec. 8, Lists})$$

$$= \frac{n!}{(n-k)!} \quad (\text{Rewrite using factorial notation})$$

2. Now if order doesn't matter, we

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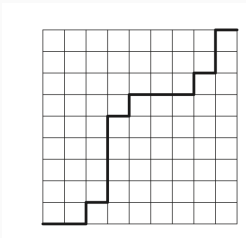
2. Now if order doesn't matter, we need to treat all reorderings of the k items as equivalent. That is, we want to count equivalence classes, each of which has $k!$ members. So by Scheinerman Theorem 16.6 (Counting equivalence classes), the total number of equivalence classes is $\frac{(n)_k}{k!}$.

Hence,

$$\binom{n}{k} = \frac{(n)_k}{k!} = \frac{n!}{k!(n-k)!}$$

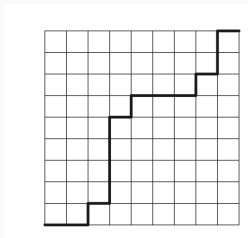
Application: Counting Lattice Paths

Problem. Suppose we want to count the number of grid paths from the lower left corner to the upper right corner in which each step of the path either goes one unit to the right or one unit vertically. How many paths are there?



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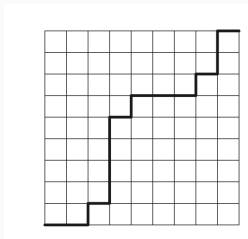
Solution. Each possible path must take 18 steps - 9 rights and 9 ups. Using R to denote right and U to denote up, one possible path is

RRU RUU UUR URR RUR UUR

Now we can count the number of paths by specifying which 9 of the 18 positions have R's. The solution is

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$$\binom{18}{9} = 48,620$$

Group exercises

Ashley, Isabelle: 6	Hager, Nathan: 12	Putnam, John: 2
Bailey, Tanna: 4	Horner, Jayden: 7	Ramos, Vydel: 13
Blaisdell, Olyver: 1	Johnson, Jaret: 10	Rathbun, Jedidiah: 13
Brown, Ashton: 12	Kintzel, Samantha: 8	Roberts, Matthew: 11
Brown, Jonathan: 6	Lameres, Alexis: 12	Roller, Nathan: 2
Cardenas, Dominic: 5	Lee, Holden: 3	Ryan, Otis: 9
Cutler, George: 5	Mayala, Reece: 3	Shevtsov, Timothy: 1
Fabik, Roland: 10	McLean, Connor: 4	Sinclair, Aidan: 6
Fellows, Audrey: 8	Mcbride, Daniel: 1	Smith, Charles: 5
Fitzgerald, Ryan: 11	Miller, Maverick: 2	Stensland, Emma: 7
Fuller, Ryan: 4	Moss, Emily: 8	Thiessen, Logan: 10
Griffen, Clara: 11	Mousseau, Matthew: 13	Thompson, Jaiden: 7
Grutsch, Tanner: 9	Neuman, Evan: 9	Vath, Ethan: 3

Group exercises

1. For each question below, determine the answer in three different ways. First, write out all the possible subsets explicitly. Second, use a proposition or theorem from the text. Third, use a different proposition or theorem from the text.
 - a. Determine the number of 2 element subsets of $\{1, 2, 3, 4, 5, 6\}$.
 - b. Determine the number of 4 element subsets of $\{1, 2, 3, 4, 5, 6\}$.

2. Scheinerman Prop 17.7 says: Let $n, k \in \mathbb{N}$ with $0 \leq k \leq n$. Then

$$\binom{n}{k} = \binom{n}{n-k}.$$

Prove this proposition using Scheinerman Theorem 17.12, which says

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

3. Fifty runners compete in a 5K race. How many different outcomes are possible? Find different answers according to the context. (a) We want to know in what place every runner finished. (b) The race is a qualifying race, and we just want to pick the ten fastest runners. (c) The race is an Olympic final event, and we care only about who gets the gold, silver, and bronze medals.
4. At a large community festival, there's a group of 30 volunteers — 8 from Missoula and 22 from other towns. The organizers want to form a five-person team to plan the main event. However, they insist that at least one person from Missoula must be on the team to make sure the town's voice is heard. How many five-person teams can be formed containing at least one person from Missoula?

Solution to group exercise #1a

There are 15 2-elements subsets of $\{1, 2, 3, 4, 5, 6\}$. We show this in 3 different ways.

1. Writing out all the subsets explicitly, we have

$$\begin{aligned}\{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{1, 6\} \\ \{2, 3\}, \{2, 4\}, \{2, 5\}, \{2, 6\} \\ \{3, 4\}, \{3, 5\}, \{3, 6\} \\ \{4, 5\}, \{4, 6\} \\ \{5, 6\}\end{aligned}$$

2. The number of subsets of size 2 is given by Scheinerman Prop. 17.5 as

$$\binom{n}{2} = \sum_{k=1}^{n-1} k.$$

In this case, since $n = 6$, we have $\binom{6}{2} = 1 + 2 + 3 + 4 + 5 = 15$. (Note that the layout of the solution to #1 suggests why the formula is true.)

3. The computational formula for $\binom{n}{k}$ (Scheinerman Theorem 17.12) is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Substituting $n = 6$ and $k = 2$, we find

$$\binom{6}{2} = \frac{6!}{2!4!} = \frac{6 \cdot 5}{2 \cdot 1} = 15.$$

Solution to group exercise #1b

There are 15 4-elements subsets of $\{1, 2, 3, 4, 5, 6\}$. We show this in 3 different ways.

1. Writing out all the subsets explicitly, we have

$$\begin{aligned} &\{3, 4, 5, 6\}, \{2, 4, 5, 6\}, \{2, 3, 5, 6\}, \{2, 3, 4, 6\}, \{2, 3, 4, 5\} \\ &\{1, 4, 5, 6\}, \{1, 3, 5, 6\}, \{1, 3, 4, 6\}, \{1, 3, 4, 5\} \\ &\{1, 2, 5, 6\}, \{1, 2, 4, 6\}, \{1, 2, 4, 5\} \\ &\{1, 2, 3, 6\}, \{1, 2, 3, 5\} \\ &\{1, 2, 3, 4\} \end{aligned}$$

2. Scheinerman Prop. 17.7 says that for $n, k \in \mathbb{N}$ with $0 \leq k \leq n$, we have

$$\binom{n}{k} = \binom{n}{n-k}$$

Hence $\binom{6}{4} = \binom{6}{2}$. Since we already know from group exercise #1a that $\binom{6}{2} = 15$, we must have $\binom{6}{4} = 15$. (Indeed, the enumeration of subsets above was obtained by taking the complement of the enumerated subsets from the previous slide.)

3. The computational formula for $\binom{n}{k}$ (Scheinerman Theorem 17.12) is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Substituting $n = 6$ and $k = 4$, we find

$$\binom{6}{2} = \frac{6!}{4!2!} = \frac{6 \cdot 5}{2 \cdot 1} = 15.$$

Solution to group exercise #2

Problem. Scheinerman Prop 17.7 says: Let $n, k \in \mathbb{N}$ with $0 \leq k \leq n$.

Then

$$\binom{n}{k} = \binom{n}{n-k}.$$

Prove this proposition using Scheinerman Theorem 17.12.

Solution. Scheinerman Theorem 17.12 says

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

If we apply Scheinerman Theorem 17.12 to the problem of finding subsets of size $n - k$ from a set of size n , we have

$$\binom{n}{n-k} = \frac{n!}{(n-k)!(n-(n-k))!} = \frac{n!}{(n-k)!k!} = \frac{n!}{k!(n-k)!}.$$

Hence,

$$\binom{n}{k} = \binom{n}{n-k}.$$

Solution to group exercise #3

Problem. Fifty runners compete in a 5K race. How many different outcomes are possible? Find different answers according to the context.

(a) We want to know in what place every runner finished. (b) The race is a qualifying race, and we just want to pick the ten fastest runners. (c) The race is an Olympic final event, and we care only about who gets the gold, silver, and bronze medals.

Solution.

- a. $50!$. Here we want the number of permutations of the 50 runners; that is the number of different ways to order them.
- b. Assuming that we don't care about the order in which the 10 runners finished, the answer is $\binom{50}{10}$. Here we want to find the number of subsets of size 10 from a pool of size 50. (If we do care about the order in which the 10 runners finished, the answer is $50 \times 49 \times \cdots 41$.)
- c. $50 \times 49 \times 48$. There are 50 choices for the first medal, which leaves 49 choices for the second medal and 48 choices for the third medal.

Solution to group exercise #4

Problem. At a large community festival, there's a group of 30 volunteers — 8 from Missoula and 22 from other towns. The organizers want to form a five-person team to plan the main event. However, they insist that at least one person from Missoula must be on the team to make sure the town's voice is heard. How many five-person teams can be formed containing at least one person from Missoula?

Solution. Let T be the total number of five-person teams. Let M be the number of five-person teams with at least one person from Missoula. Let N be the number of five-person teams can with no people from Missoula. Then

$$M = T - N.$$

Now $T = \binom{30}{5}$ by definition of the binomial coefficient. (Here we can choose our five-person team from any of the 30 volunteers). Similarly, $N = \binom{22}{5}$. (Here we must choose our five-person team solely from the 22 volunteers aren't from Missoula). Hence,

$$M = \binom{30}{5} - \binom{22}{5}.$$